RECOVERING AN OBJECT AS THE DERIVED LIMIT OF ITS TRUNCATIONS UNDER A VANISHING HYPOTHESIS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the matching of hypotheses and the application are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Truncations and the derived limit of the system of truncations. Same chapter together with the chapter Derived Categories, the map from an object to the derived limit of its truncations. Supplies the map whose being an isomorphism is asserted.
- Vanishing criterion for cohomology over a covering family. Same chapter, the companion criterion identifying the codomain of the map with the object itself under vanishing hypotheses of the shape appearing in the statement. Supplies the criterion applied in the proof.

Source reference. Original remark 0H72: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5407–5407. Navigation pointer invoked at proof line 5407; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6M: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5320–5320. Navigation pointer invoked at proof line 5412; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $E \in D(\mathcal{O})$. Assume there exists a function $p(-) : \mathbf{Z} \to \mathbf{Z}$ and a subset $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ such that*

(1) *every object of $\mathcal{C}$ has a covering whose members are elements of $\mathcal{B}$,*
(2) *$H^p(V, H^{m-p}(E)) = 0$ for $p > p(m)$ and $V \in \mathcal{B}$.*

Then the map $E \to R\lim \tau_{\geq -n} E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O})$.

Proof. Apply Lemma 0D6M with $p(V, m) = p(m)$ and Cov_V equal to the set of coverings $\{V_i \to V\}$ with $V_i \in \mathcal{B}$ for all i. □